

CSC 361

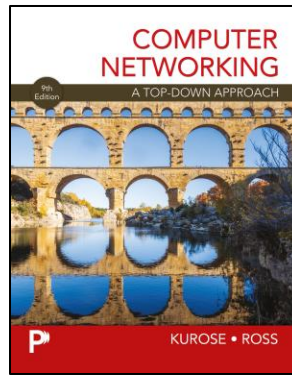
Computer Networks

Network Layer

(Chapter 4 – Data Plane)

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Summer 2026



Network Layer: Our Goals

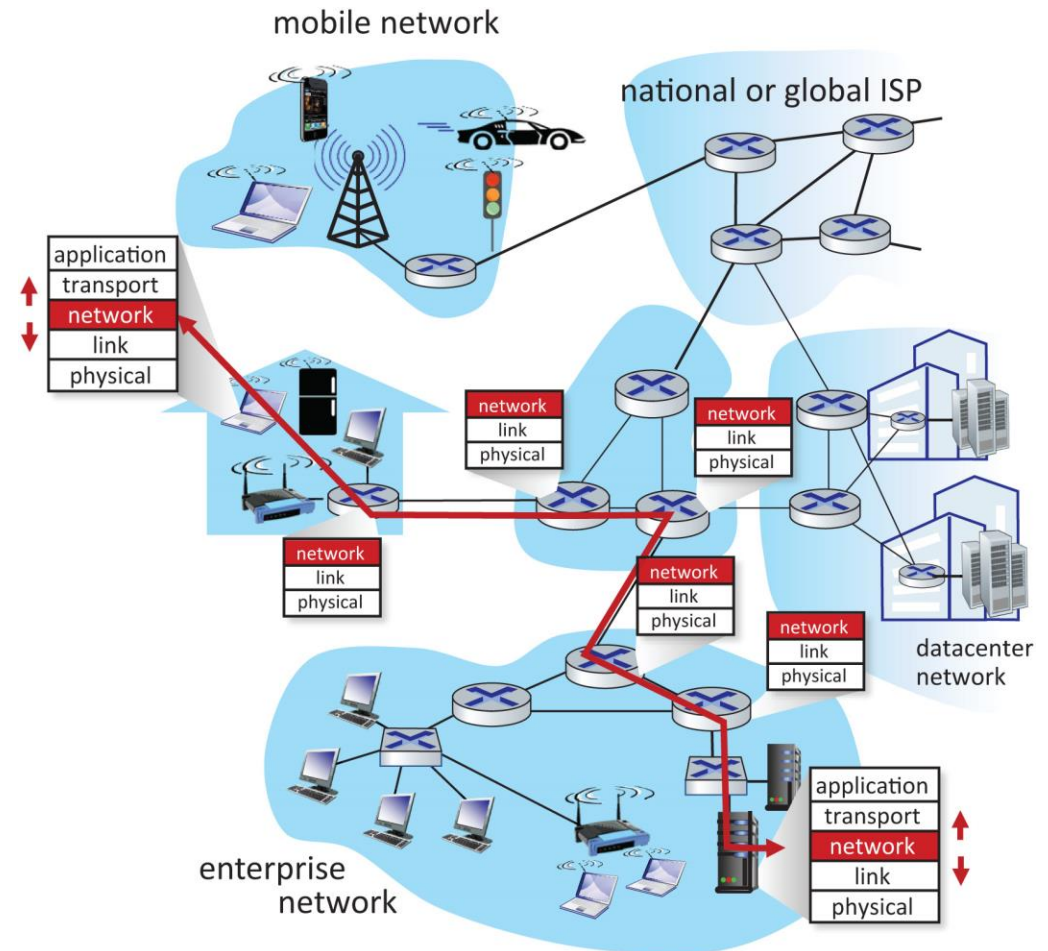
- understand principles behind network layer services, focusing on data plane:
 - network layer service models
 - forwarding versus routing
 - how a router works
 - addressing
 - generalized forwarding
 - Internet architecture
- instantiation, implementation in the Internet
 - IP protocol
 - NAT, middleboxes

Network Layer: “Data plane” Roadmap (1 of 6)

- Network layer: overview
 - data plane
 - control plane
- What’s inside a router
 - input/output ports, switching
 - buffer management
 - scheduling
- IP: the Internet Protocol
 - datagram format, addressing
 - network address translation (NAT)
 - IPv6
- Generalized Forwarding
 - Match+action
 - OpenFlow
 - Middleboxes
- Internet architecture

Network-Layer Services and Protocols

- transport segment from sending to receiving host
 - **sender**: encapsulates segments into datagrams, passes to link layer
 - **receiver**: delivers segments to transport layer protocol
- network layer protocols in **every Internet device**: hosts, routers
- **routers**:
 - examines header fields in all IP datagrams passing through it
 - moves datagrams from input ports to output ports to transfer datagrams along end-end path



Two key Network-Layer Functions

network-layer functions:

- **forwarding:** move packets from a router's input link to appropriate router output link
- **routing:** determine route taken by packets from source to destination
 - routing algorithms

analogy: taking a trip

- **forwarding:** process of getting through single interchange
- **routing:** process of planning trip from source to destination



forwarding



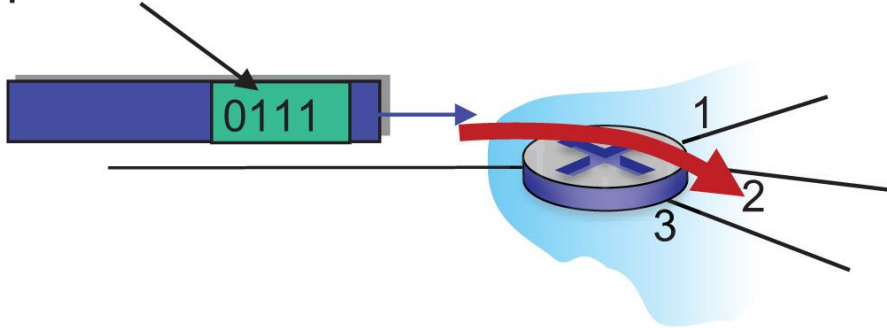
routing

Network Layer: Data Plane, Control Plane

Data plane:

- **local**, per-router function
- determines how datagram arriving on router input port is forwarded to router output port

values in arriving
packet header

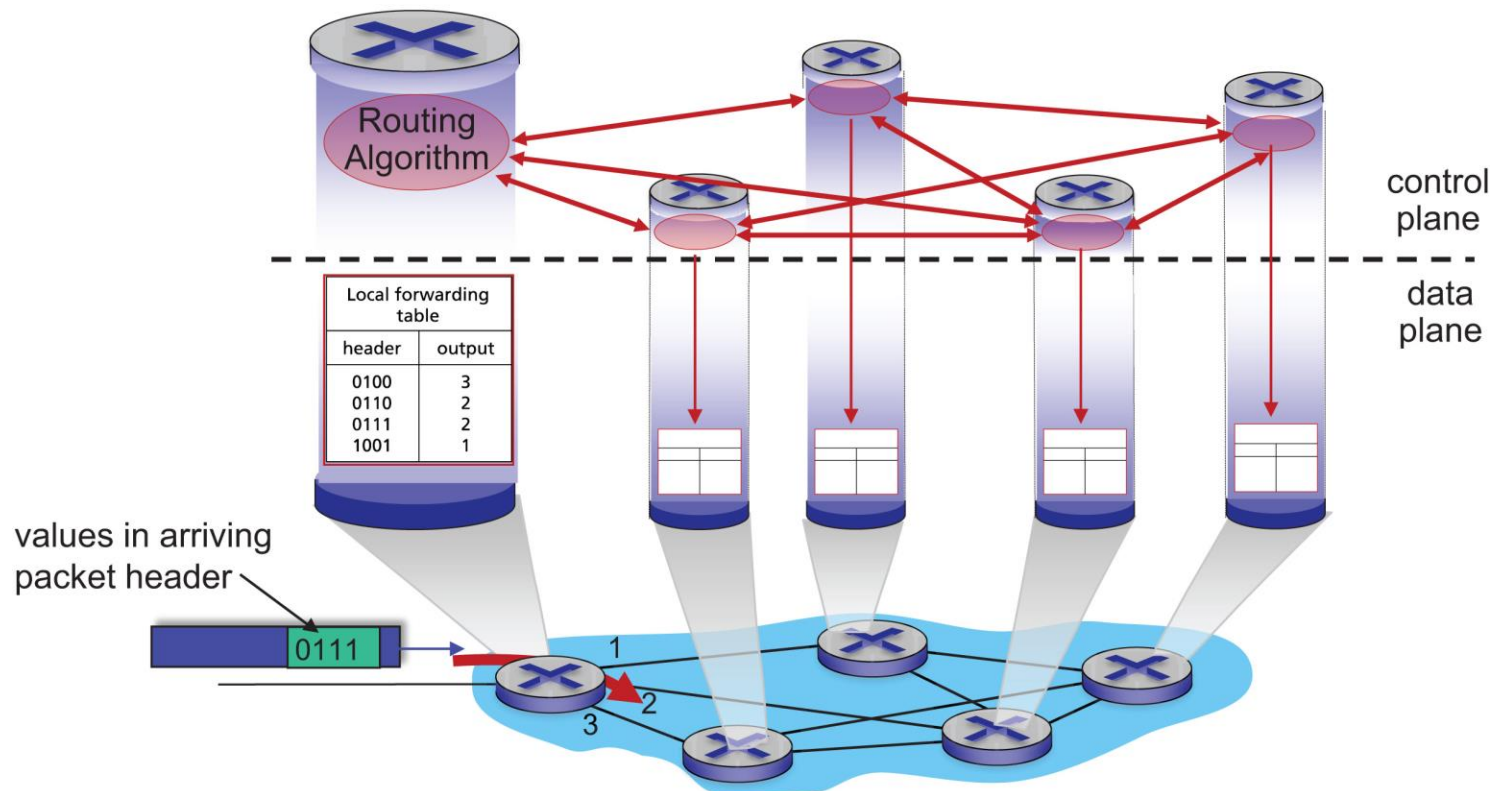


Control plane

- **network-wide** logic
- determines how datagram is routed among routers along end-end path from source host to destination host
- two control-plane approaches:
 - **traditional routing algorithms:** implemented in routers
 - **software-defined networking (SDN):** implemented in (remote) servers

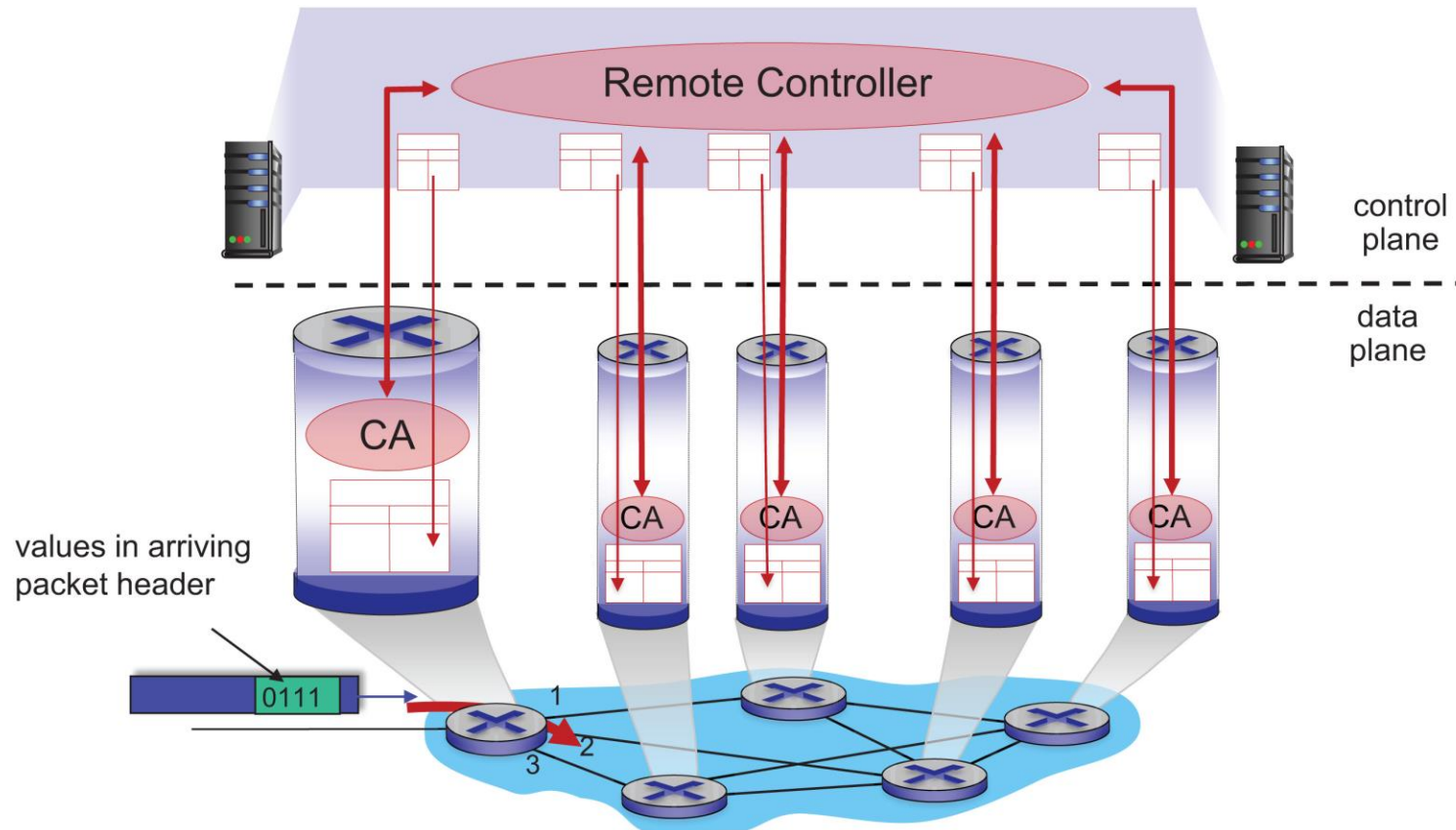
Per-Router Control Plane

Individual routing algorithm components **in each and every router** interact in the control plane



Software-Defined Networking (SDN) Control Plane

Remote controller computes, installs forwarding tables in routers



Network Service Model

Q: What **service model** for “channel” transporting datagrams from sender to receiver?

example services for **individual** datagrams:

- guaranteed delivery
- guaranteed delivery with less than 40 msec delay

example services for a **flow** of datagrams:

- in-order datagram delivery
- guaranteed minimum bandwidth to flow
- restrictions on changes in inter-packet spacing

Network-Layer Service Model (1 of 2)

Quality of Service (QoS) Guarantees ?

Network Architecture	Service Model	Bandwidth	Loss	Order	Timing
Internet	best effort	none	no	no	no

Internet “best effort” service model

No guarantees on:

- i. successful datagram delivery to destination
- ii. timing or order of delivery
- iii. bandwidth available to end-end flow

Network-Layer Service Model (2 of 2)

Quality of Service (QoS) Guarantees ?

Network Architecture	Service Model	Bandwidth	Loss	Order	Timing
Internet	best effort	none	no	no	no
ATM	Constant Bit Rate	Constant rate	yes	yes	yes
ATM	Available Bit Rate	Guaranteed min	no	yes	no
Internet	Intserv Guaranteed (RFC 1633)	yes	yes	yes	yes
Internet	Diffserv (RFC 2475)	possible	possibly	possibly	no

Reflections on Best-Effort Service:

- simplicity of mechanism has allowed Internet to be widely deployed adopted
- sufficient provisioning of bandwidth allows performance of real-time applications (e.g., interactive voice, video) to be “good enough” for “most of the time”
- replicated, application-layer distributed services (datacenters, content distribution networks) connecting close to clients’ networks, allow services to be provided from multiple locations
- congestion control of “elastic” services helps

It's hard to argue with success of best-effort service model

Network Layer: “Data plane” Roadmap (2 of 6)

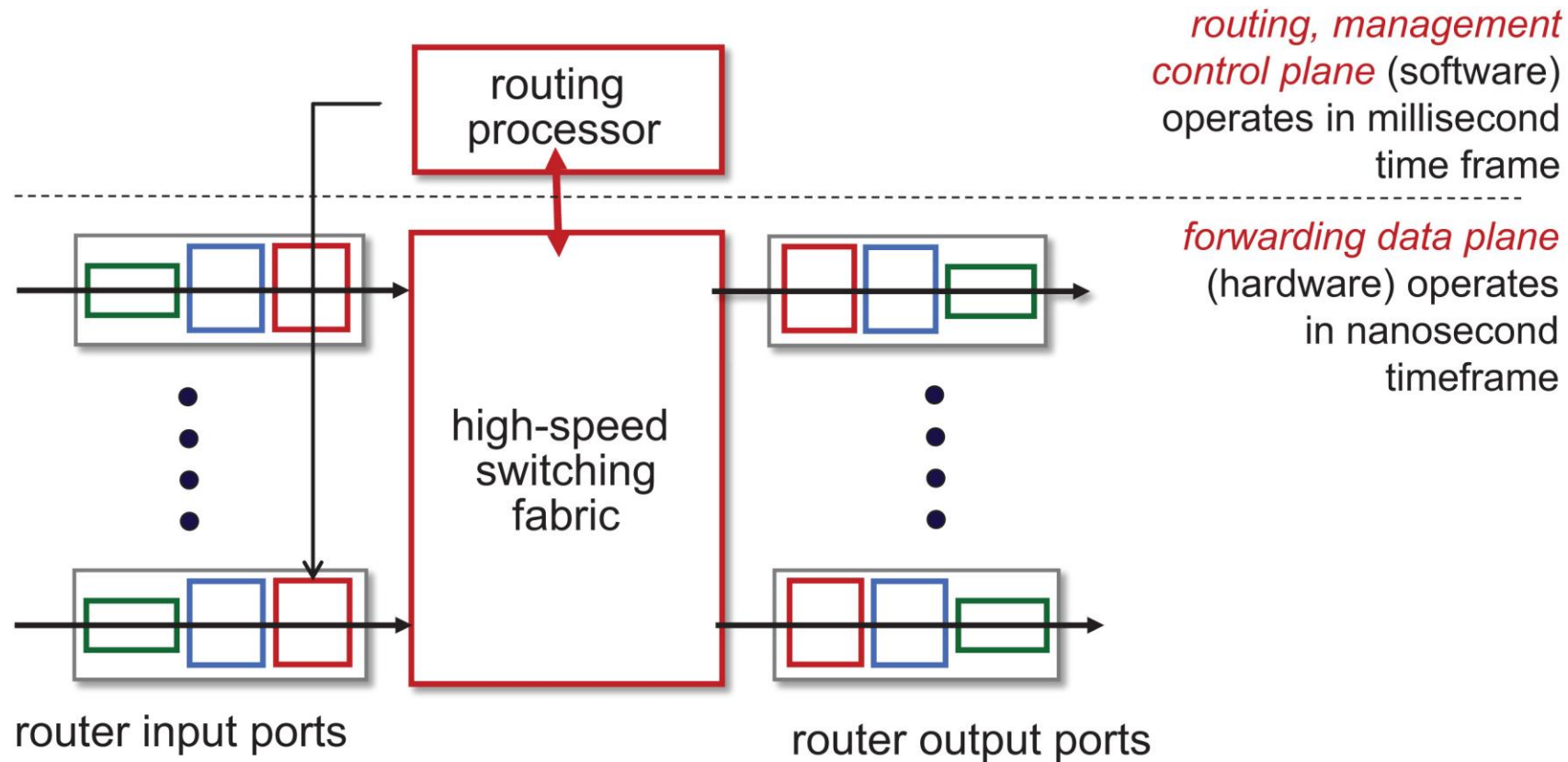
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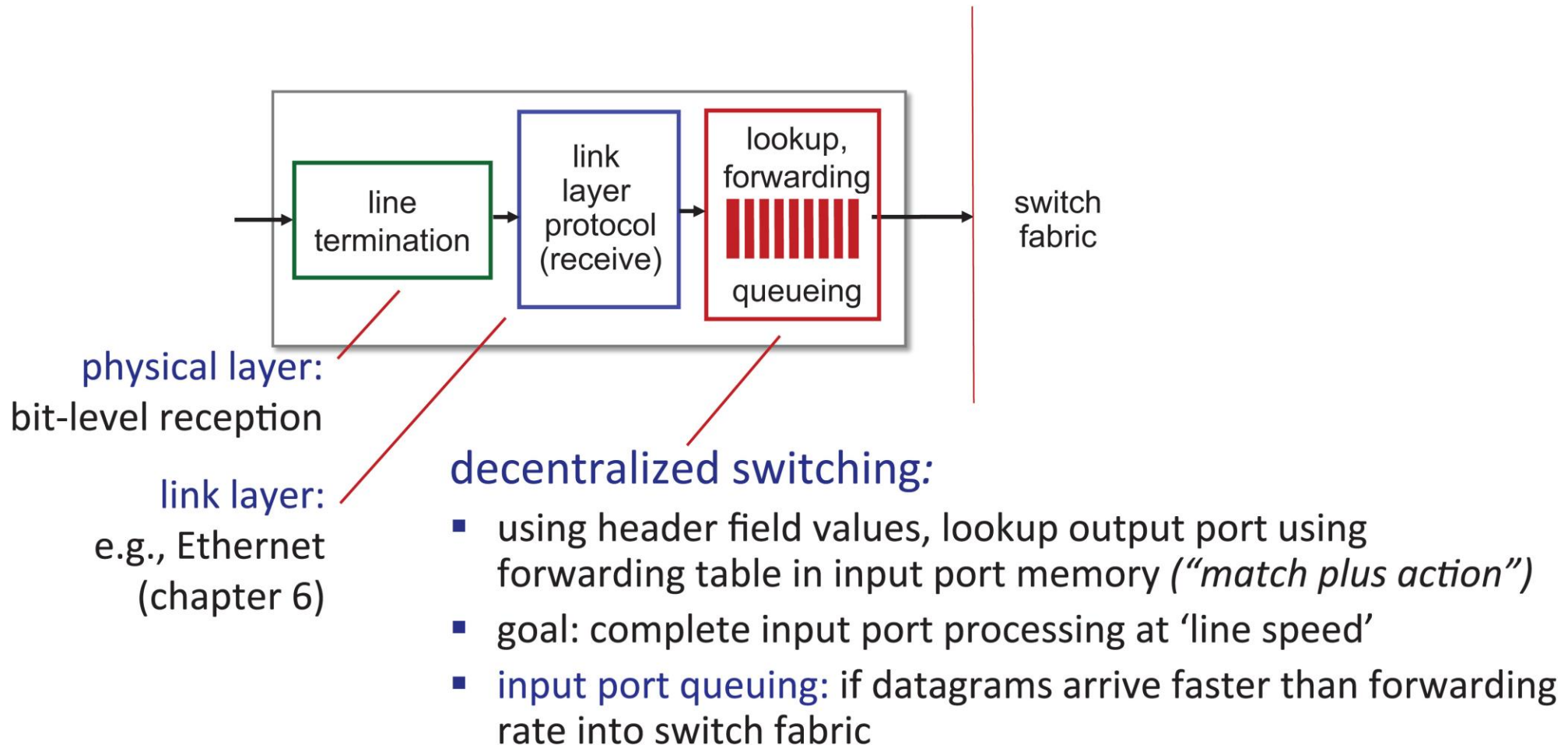
- Generalized Forwarding
 - Match+action
 - OpenFlow
 - Middleboxes
- Internet architecture

Router Architecture Overview

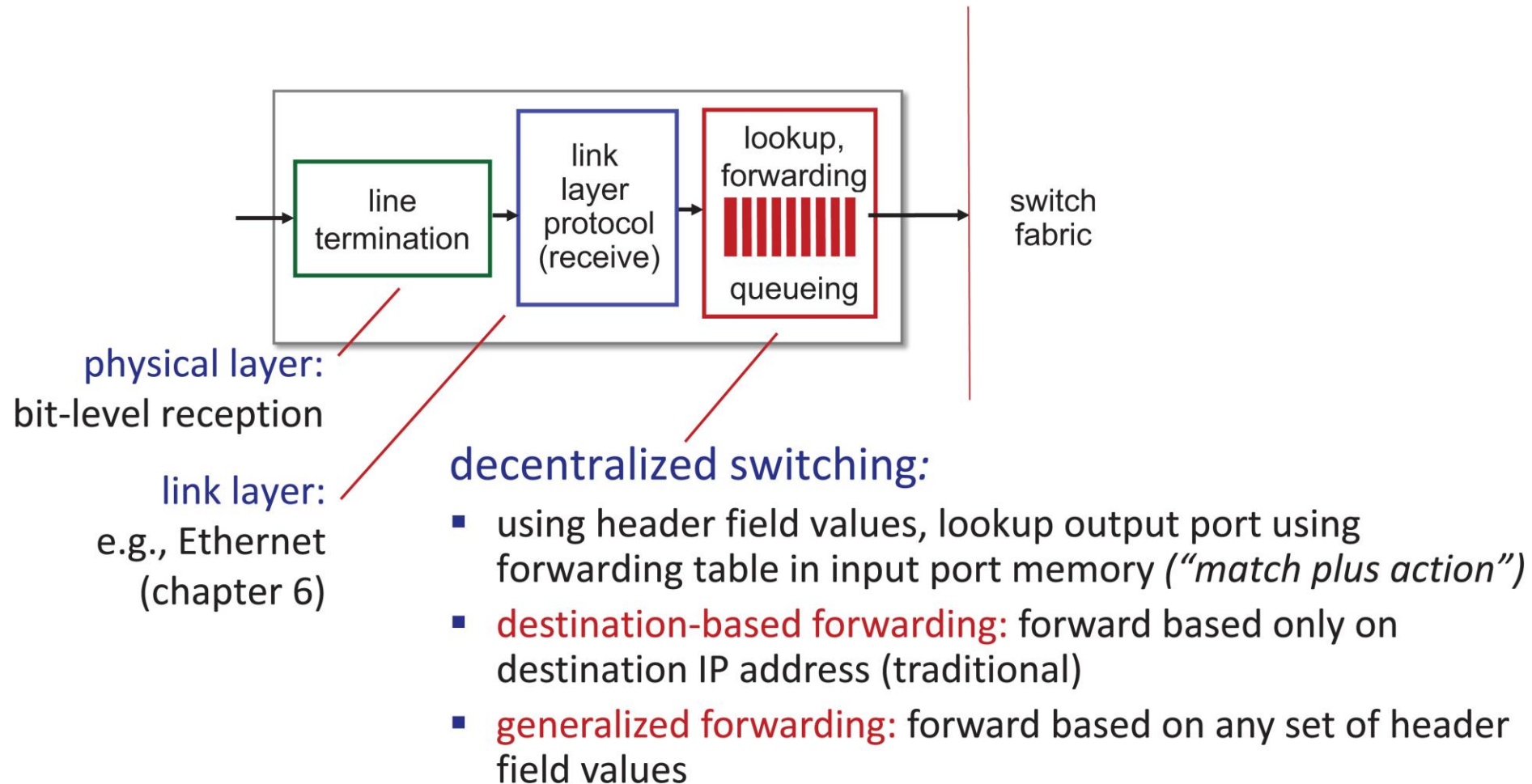
high-level view of generic router architecture:



Input Port Functions (1 of 2)



Input Port Functions (2 of 2)



Destination-Based Forwarding

<i>forwarding table</i>	
Destination Address Range	Link Interface
11001000 00010111 00010000 00000000 through 11001000 00010111 00010000 00000100 through 11001000 00010111 00010000 00000111 11001000 00010111 00011000 11111111	0 3
11001000 00010111 00011001 00000000 through 11001000 00010111 00011111 11111111	2
otherwise	3

Q: but what happens if ranges don't divide up so nicely?

Longest Prefix Matching (1 of 5)

longest prefix match

when looking for forwarding table entry for given destination address, use **longest** address prefix that matches destination address.

Destination Address Range				Link interface
11001000	00010111	00010***	*****	0
11001000	00010111	00011000	*****	1
11001000	00010111	00011***	*****	2
otherwise				3

examples:

11001000	00010111	00010110	10100001	which interface?
11001000	00010111	00011000	10101010	which interface?

Longest Prefix Matching (2 of 5)

longest prefix match

when looking for forwarding table entry for given destination address, use **longest** address prefix that matches destination address.

Destination Address Range	Link interface
11001000 00010111 00010*** *****	0
11001000 00010111 00011000 *****	1
11001000 00010111 00011*** *****	2
otherwise	3

examples:

11001000 00010111 00010110 10100001 which interface?
11001000 00010111 00011000 10101010 which interface?

Longest Prefix Matching (3 of 5)

longest prefix match

when looking for forwarding table entry for given destination address, use **longest** address prefix that matches destination address.

Destination Address Range	Link interface
11001000 00010111 00010*** *****	0
11001000 00010111 00011000 *****	1
11001000 00010111 00011*** *****	2
otherwise	3

examples:

match!

11001000 00010111 00010110 10100001 which interface?

11001000 00010111 00011000 10101010 which interface?

Longest Prefix Matching (4 of 5)

longest prefix match

when looking for forwarding table entry for given destination address, use **longest** address prefix that matches destination address.

Destination Address Range				Link interface
11001000	00010111	00010***	*****	0
11001000	00010111	00011000	*****	1
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otherwise				3

match!

examples:

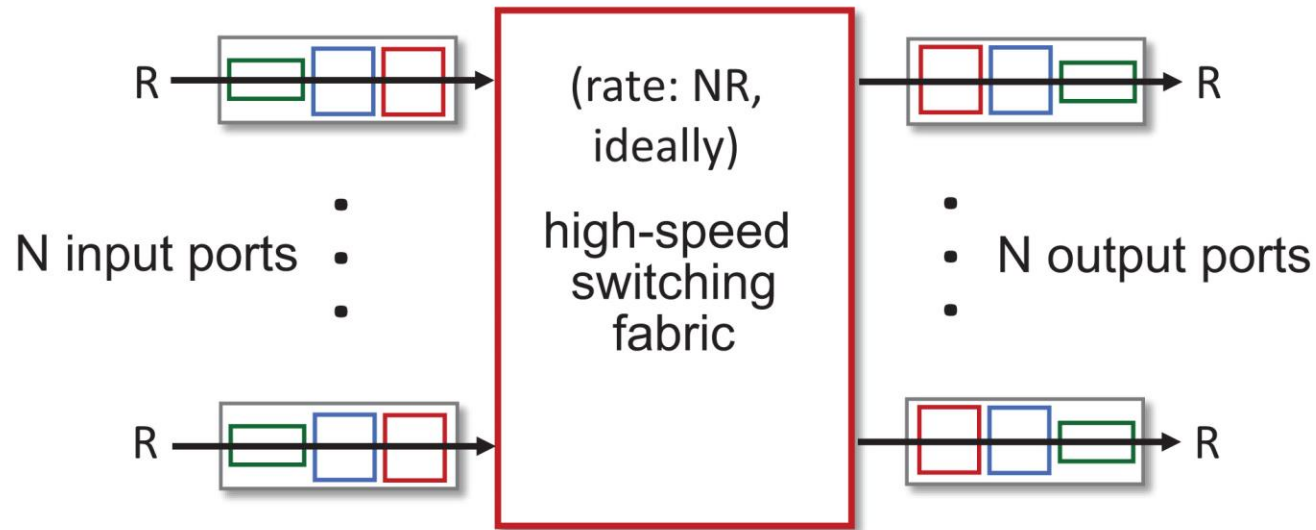
11001000	00010111	00010110	10100001	which interface?
11001000	00010111	00011000	10101010	which interface?

Longest Prefix Matching (5 of 5)

- we'll see **why** longest prefix matching is used shortly, when we study addressing

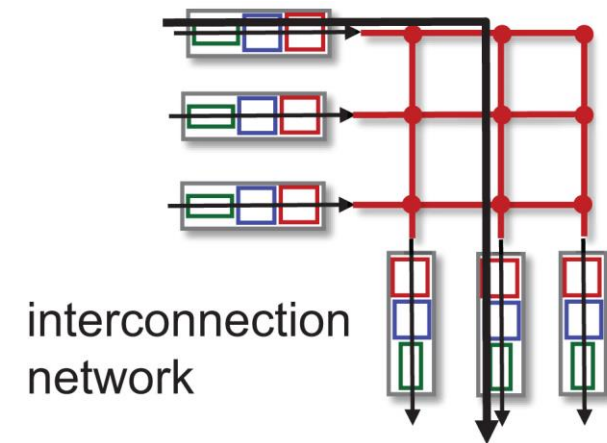
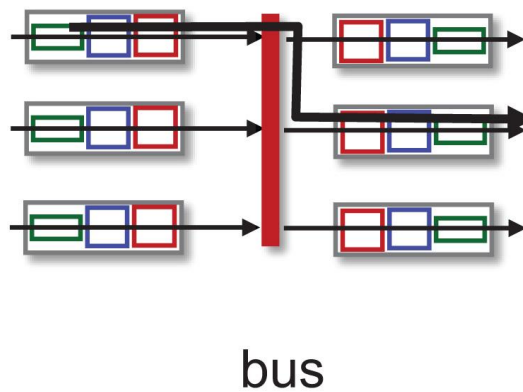
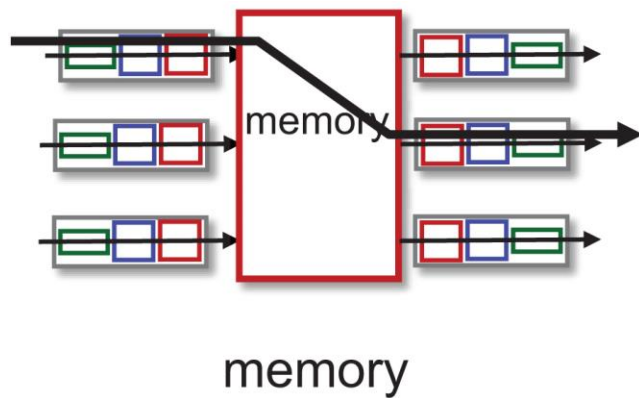
Switching Fabrics (1 of 2)

- transfer packet from input link to appropriate output link
- **switching rate**: rate at which packets can be transfer from inputs to outputs
 - often measured as multiple of input/output line rate
 - N inputs: switching rate N times line rate desirable



Switching Fabrics (2 of 2)

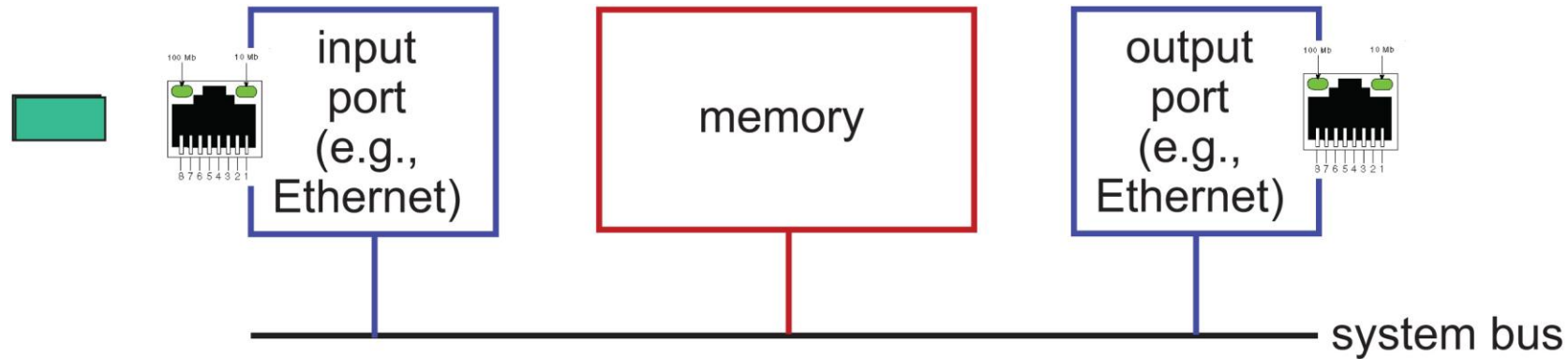
- transfer packet from input link to appropriate output link
- **switching rate**: rate at which packets can be transfer from inputs to outputs
 - often measured as multiple of input/output line rate
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- three major types of switching fabrics:



Switching Via Memory

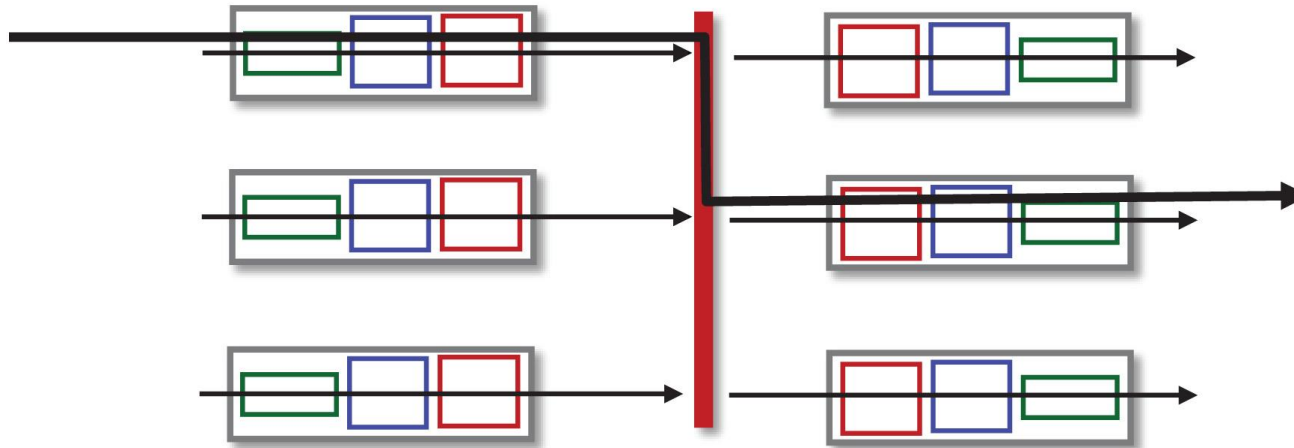
first generation routers:

- traditional computers with switching under direct control of CPU
- packet copied to system's memory
- speed limited by memory bandwidth (2 bus crossings per datagram)



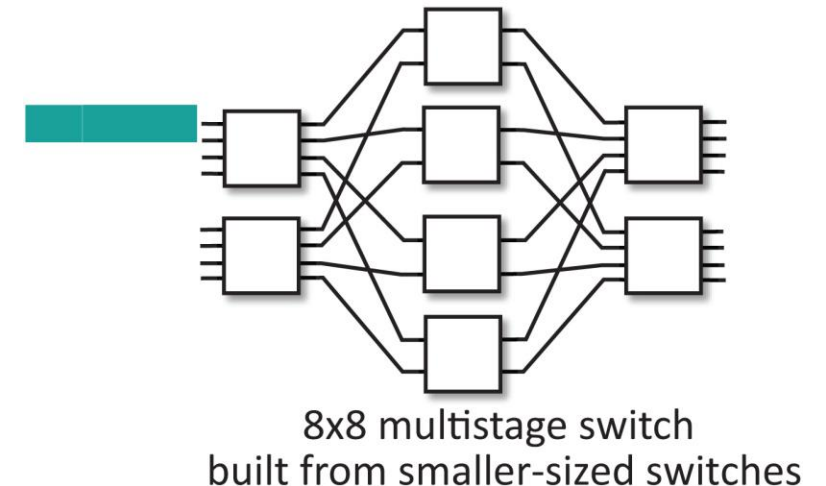
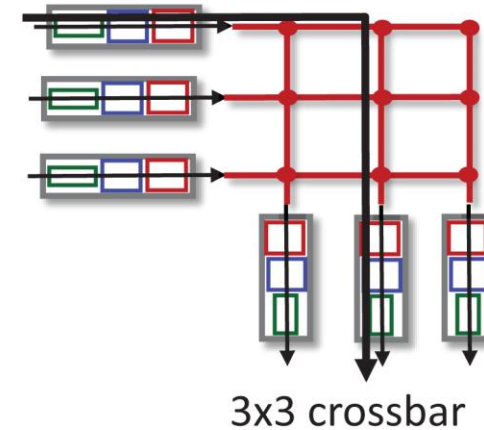
Switching Via a Bus

- datagram from input port memory to output port memory via a shared bus
- **bus contention:** switching speed limited by bus bandwidth
- 32 Gbps bus, Cisco 5600: sufficient speed for access routers



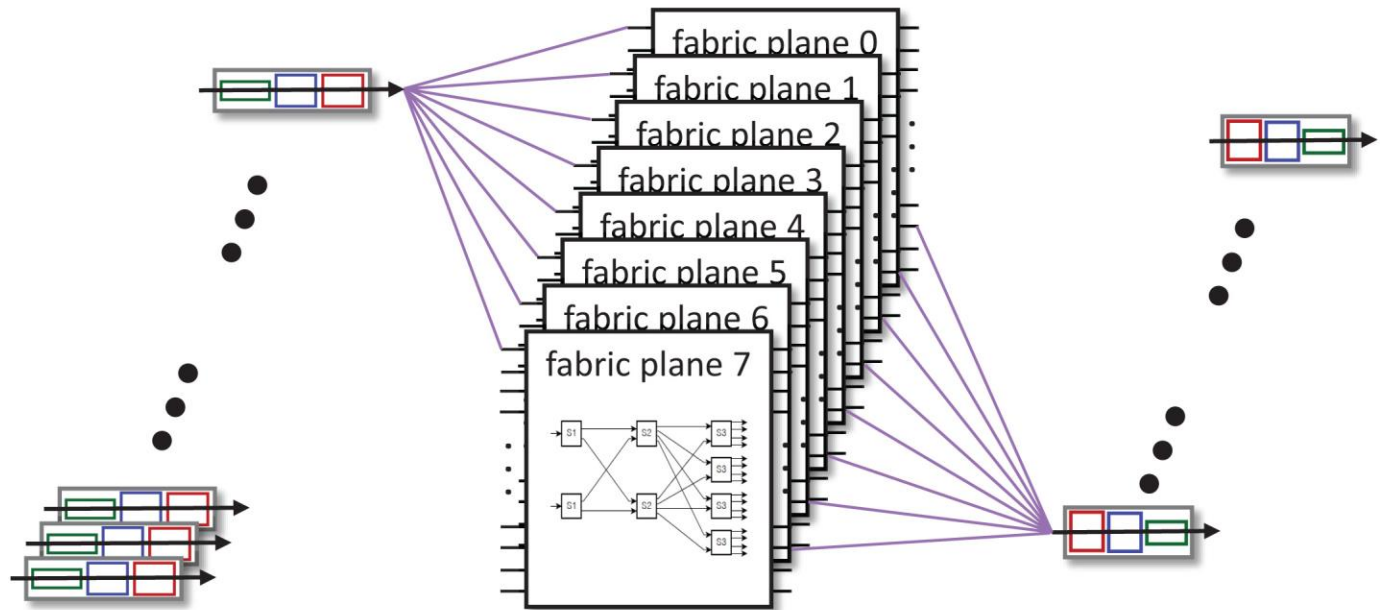
Switching Via Interconnection Network (1 of 2)

- Crossbar, Clos networks, other interconnection nets initially developed to connect processors in multiprocessor
- **multistage switch**: $n \times n$ switch from multiple stages of smaller switches
- **exploiting parallelism**:
 - fragment datagram into fixed length cells on entry
 - switch cells through the fabric, reassemble datagram at exit



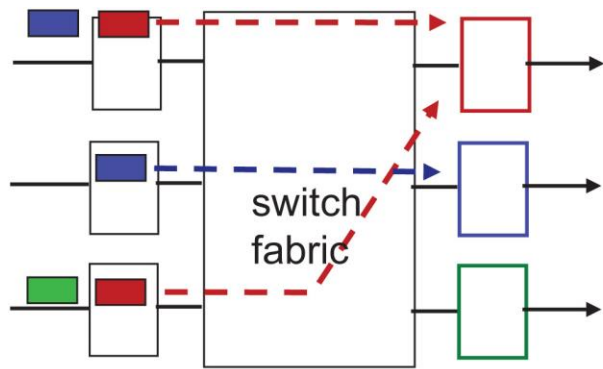
Switching Via Interconnection Network (2 of 2)

- scaling, using multiple switching “planes” in parallel:
 - speedup, scaleup via parallelism
- Cisco CRS router:
 - basic unit: 8 switching planes
 - each plane: 3-stage interconnection network
 - up to 100's Tbps switching capacity

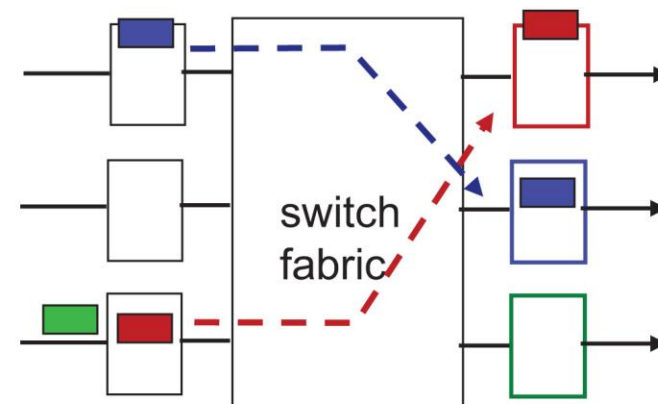


Input Port Queuing

- If switch fabric slower than input ports combined → queueing may occur at inputs
 - queueing delay and loss due to input buffer overflow!
- **Head-of-the-Line (HOL) blocking:** queued datagram at front of queue prevents others in queue from moving forward

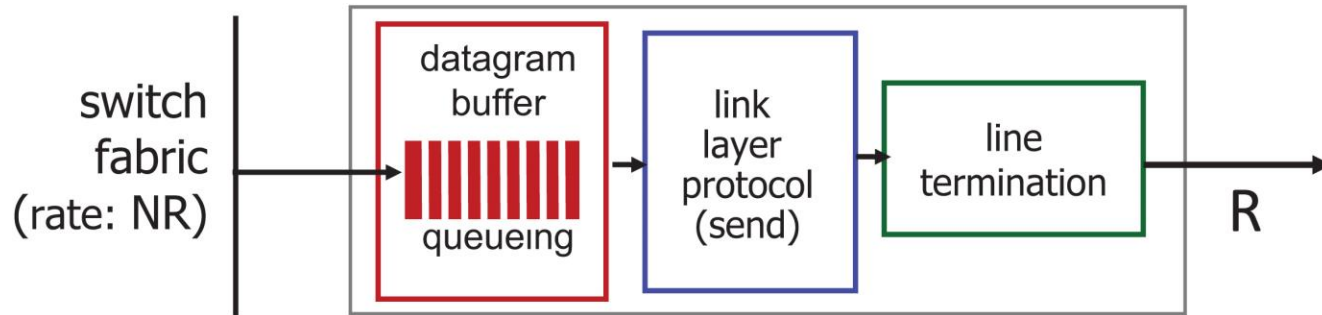


output port contention: only one red datagram can be transferred. lower red packet is *blocked*



one packet time later: green packet experiences HOL blocking

Output Port Queuing (1 of 2)



- **Buffering** required when datagrams arrive from fabric faster than link transmission rate. **Drop policy:** which datagrams to drop if no free buffers?
- **Scheduling discipline** chooses among queued datagrams for transmission

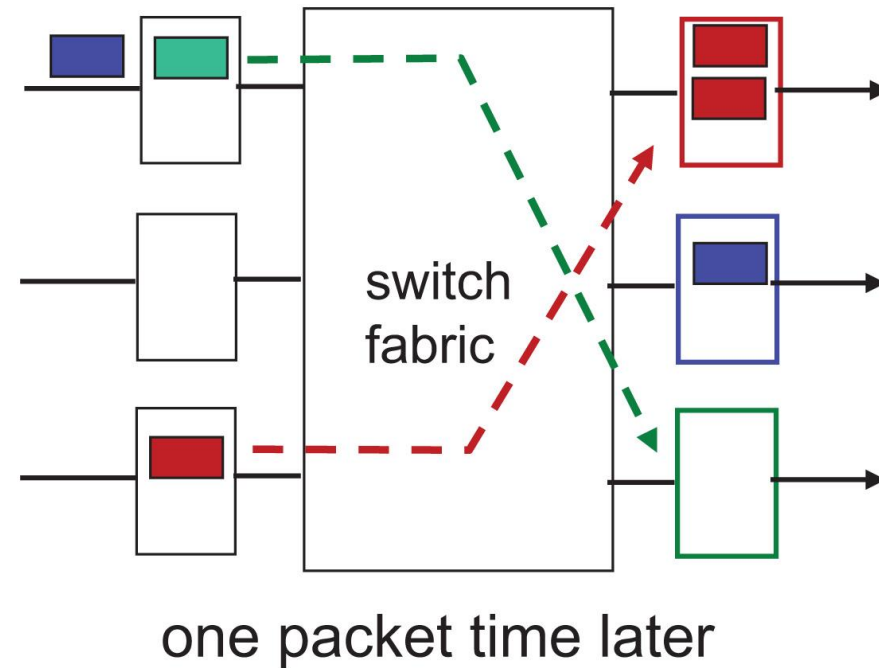
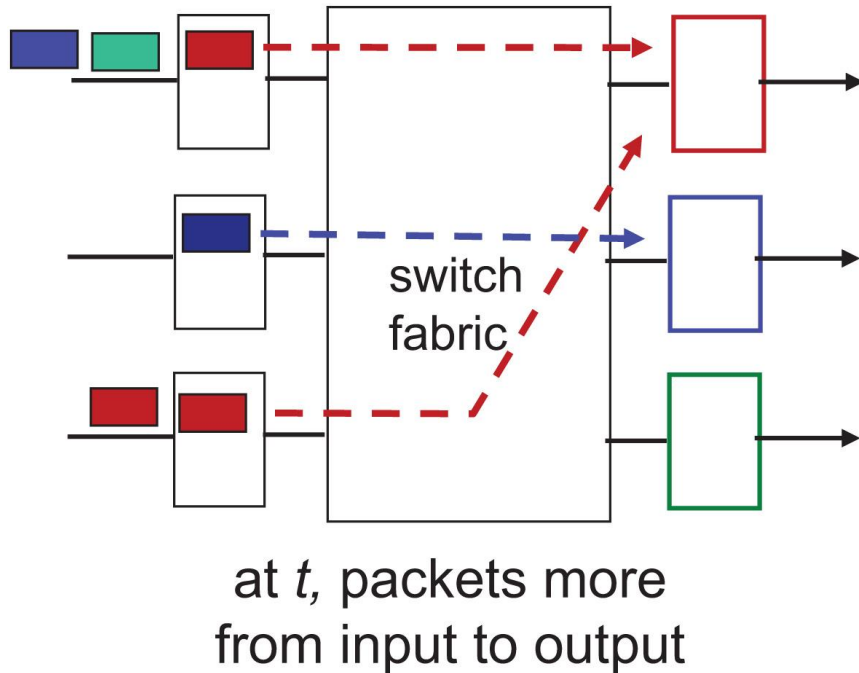


Datagrams can be lost due to congestion, lack of buffers



Priority scheduling – who gets best performance, network neutrality

Output Port Queuing (2 of 2)



- buffering when arrival rate via switch exceeds output line speed
- **queueing (delay) and loss due to output port buffer overflow!**

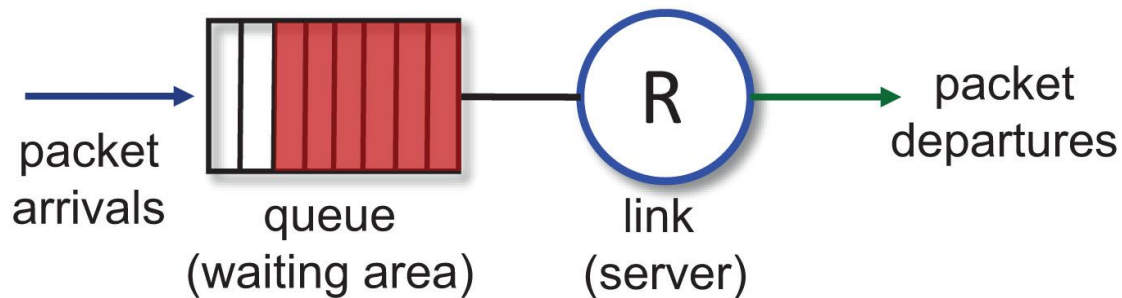
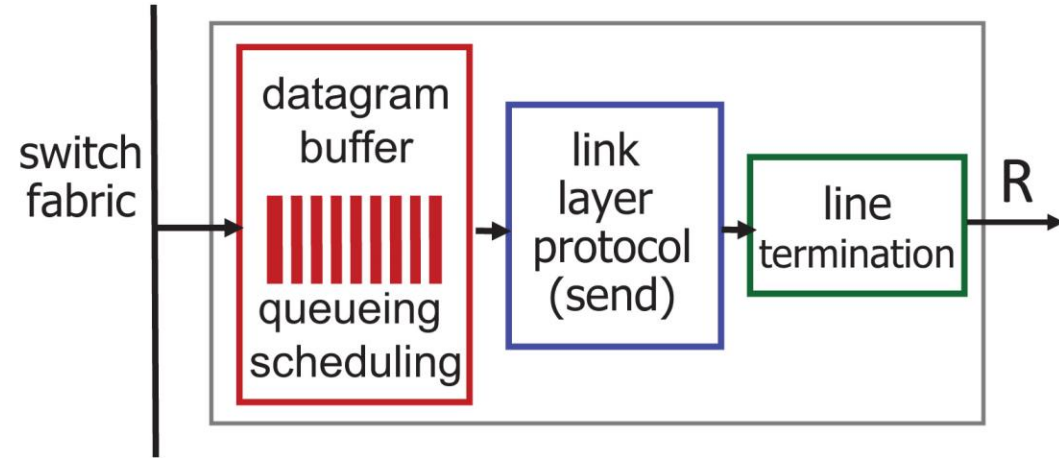
How Much Buffering?

- RFC 3439 rule of thumb: average buffering equal to “typical” RTT (say 250 msec) times link capacity C
 - e.g., $C = 10$ Gbps link: 2.5 Gbit buffer
- more recent recommendation: with N flows, buffering equal to

$$\frac{RTT_g C}{\sqrt{N}}$$

- but **too** much buffering can increase delays (particularly in home routers)
 - long RTTs: poor performance for real-time apps, sluggish TCP response
 - recall delay-based congestion control: “keep bottleneck link just full enough (busy) but no fuller”

Buffer Management



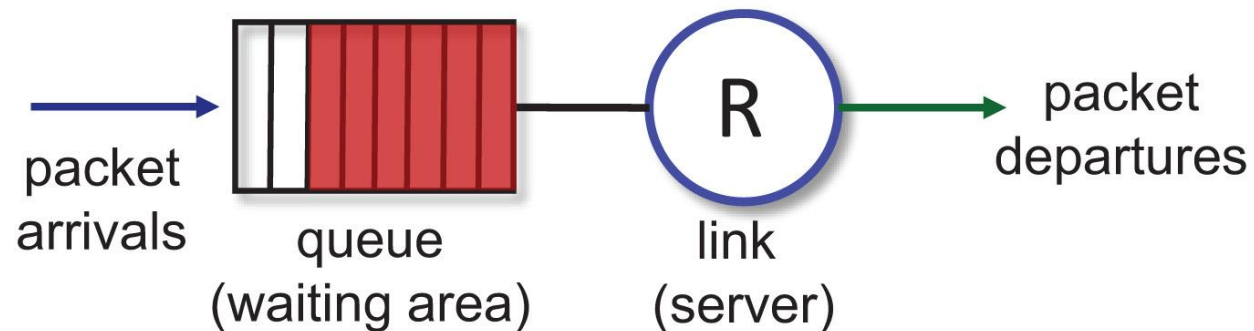
buffer management:

- **drop**: which packet to add, drop when buffers are full
 - **tail drop**: drop arriving packet
 - **priority**: drop/remove on priority basis
- **marking**: which packets to mark to signal congestion (ECN, RED)

Packet Scheduling: FCFS

- **packet scheduling**: deciding which packet to send next on link
 - first come, first served
 - priority
 - round robin
 - weighted fair queueing
- **FCFS**: packets transmitted in order of arrival to output port
 - also known as: First-in-first-out (FIFO)
 - real world examples?

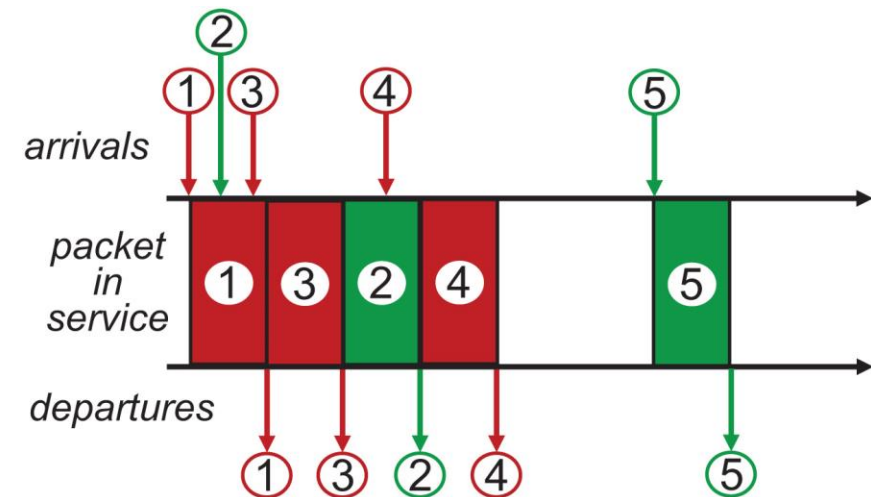
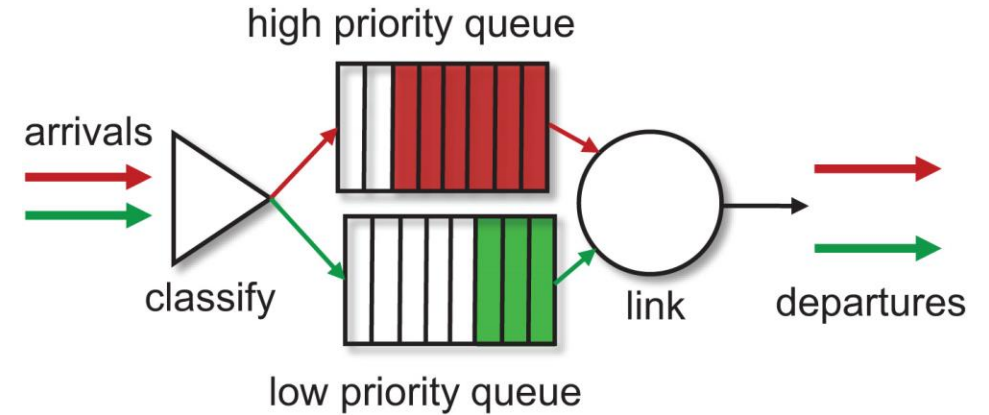
Abstraction: queue



Scheduling Policies: Priority

Priority scheduling:

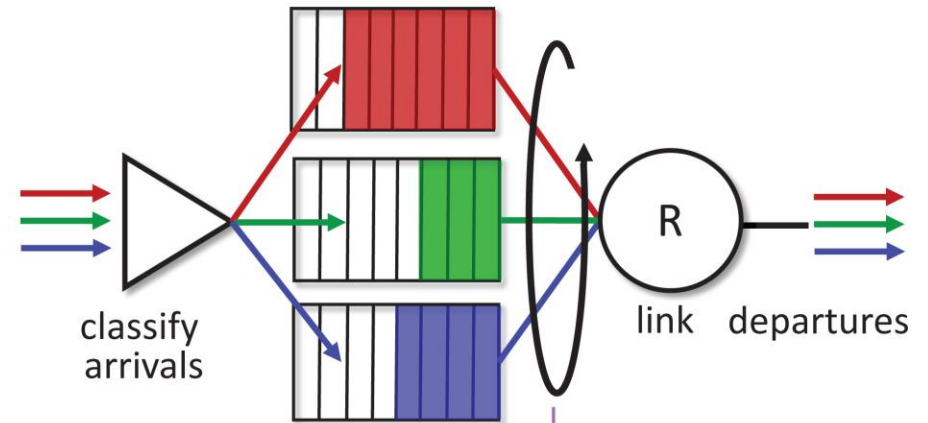
- arriving traffic classified, queued by class
 - any header fields can be used for classification
- send packet from highest priority queue that has buffered packets
 - FCFS within priority class



Scheduling Policies: Round Robin

Round Robin (RR) scheduling:

- arriving traffic classified, queued by class
 - any header fields can be used for classification
- server cyclically, repeatedly scans class queues, sending one complete packet from each class (if available) in turn



Scheduling Policies: Weighted Fair Queueing

Weighted Fair Queueing (WFQ):

- generalized Round Robin
- each class, i , has weight, w_i , and gets weighted amount of service in each cycle:

$$\frac{w_i}{\sum_j w_j}$$

- minimum bandwidth guarantee (per-traffic-class)

